Hasan Bukhari

Computational Biology • Bioinformatics • Research Engineering

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Education

University of Southern Mississippi — **Expected Dec 2027**
B.S. in Computer and Information Sciences; Minor in Biological Sciences — *Hattiesburg, MS*
GPA: 4.00 • President's List • Academic Excellence Scholar

Research & Technical Experience

Independent Computational Biology Researcher — **2026 – Present**
CancerShift and Chromatin Contact Prediction Across Chromosomes — *Independent*

- **CancerShift:** Developed a leakage-controlled multicancer classification workflow spanning TCGA, GEO, and CPTAC-GDC cohorts; achieved 99.59% accuracy on a blind microarray cohort and 98.24% on external RNA-seq data.
- Designed A17, a residual MLP predicting Hi-C contact structure from epigenomic signals and genomic-distance features under chromosome-level holdouts and a locked test set.
- Investigated cross-cohort and cross-chromosome generalization, including negative calibration experiments; document methods, limitations, and results for manuscripts in preparation.

Undergraduate Researcher, Ichthyology Lab — **Spring 2026 – Present**
University of Southern Mississippi — *Hattiesburg, MS*

- Analyze chromosome-scale *Fundulus* genome assemblies across related species to evaluate synteny, chromosome correspondence, structural variation, and candidate fusion/fission patterns.
- Build reproducible Python workflows, alignment-derived correspondence matrices, and technical reports translating genomic evidence and limitations for laboratory review.

Data Analyst Intern, Health, Safety & Environment — **Jul 2026 – Aug 2026**
Engro Technical Services — *Karachi, Pakistan*

- Built executive analytics over 130,000+ incident, near-miss, inspection, audit, waiver, and corrective-action records using Power BI, DAX, Python, and Power Query.
- Developed FastAPI/Python workflows for risk scoring and reporting and automated change monitoring across approximately 1,750 engineering and HSE files.

Software Engineering Intern — **Aug 2023 – Feb 2024**
Shahid I.T. Consultants — *Karachi, Pakistan*

- Automated recurring record-update workflows in Python, reducing manual processing by 30%; supported integration testing, deployment documentation, and application maintenance.

Selected Research Software

BRACE — 1st Place, VOXO Hatchathon 2026 Bio Track | Python, FastAPI, scikit-learn, SciPy | GitHub — **2026**

- Co-built BRACE (Biological Reliability Assessment across Cohorts and Experiments), a bioinformatics workbench for gene-signature replication and model transportability across independent cohorts, with cohort compatibility checks, shift analysis, cross-cohort evaluation, and reproducible evidence reporting.

BioLitGraph — Biomedical Literature Graph Explorer | Python, Flask, PubMed/PubTator, NetworkX | GitHub — **2026**

- Built a biomedical literature-mining application that retrieves publications, normalizes biomedical entities into evidence-linked relationship graphs, and exports structured CSV/JSON analysis data.

PhyloPlayground — Interactive Phylogenetics Toolkit | Python, Streamlit, NumPy, Plotly | GitHub — **2025**

- Implemented Neighbor-Joining and UPGMA inference, JC69/K2P evolutionary distances, FASTA parsing, bootstrap support, and interactive phylogenetic visualization.

QTL Analysis Toolkit | Python, pandas, Matplotlib | GitHub — **2025**

- Implemented LOD scans, permutation-based significance thresholds, significant-marker exports, and publication-ready statistical-genomics plots.

Technical Skills

Programming & Modeling: Python, R, SQL, C++; PyTorch, scikit-learn, pandas, NumPy, SciPy, Biopython
Computational Genomics: RNA-seq, microarrays, Hi-C, ATAC-seq, epigenomics, synteny, structural variation, QTL analysis, phylogenetics
Research Engineering: Linux/WSL, Git/GitHub, FastAPI, Flask, REST APIs, Power BI, DAX, Power Query, Excel
Data Resources: TCGA, GEO, CPTAC, ENCODE, Ensembl, PubMed/PubTator

Leadership & Honors

Leadership: President & Founder, Student Poets Association and Creative Writing Club, USM.

Selected Honors: 1st Place, VOXO Hatchathon 2026 Bio Track; Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026).